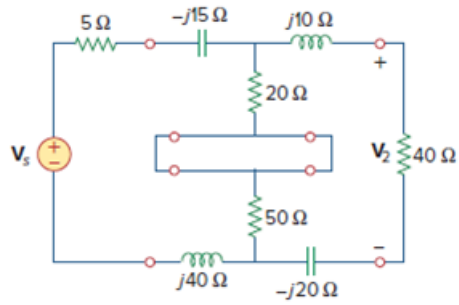


Problem

Find V_2/V_s in the circuit in Fig. 19.43.

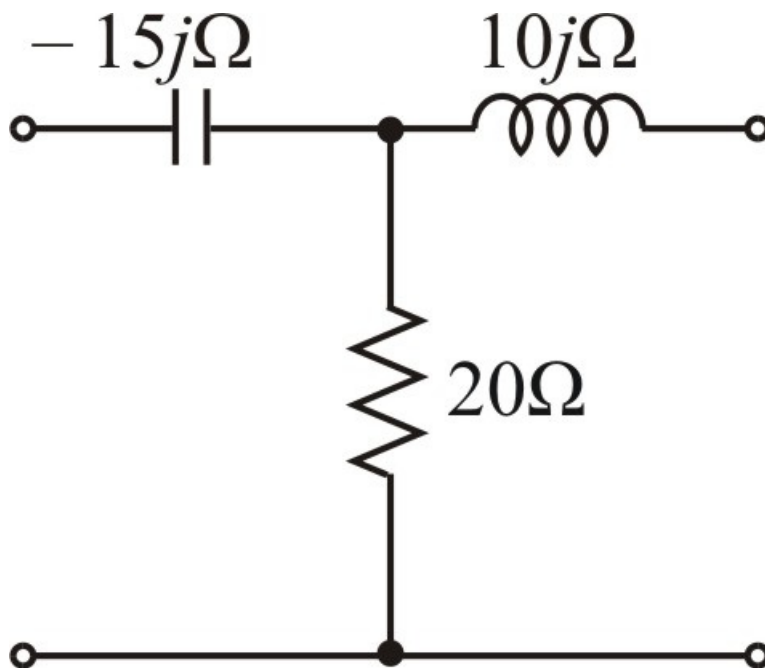
Figure 19.43



Step-by-step solution

Step 1 of 15

Consider two – port networks N_a and N_b are for N_a :-

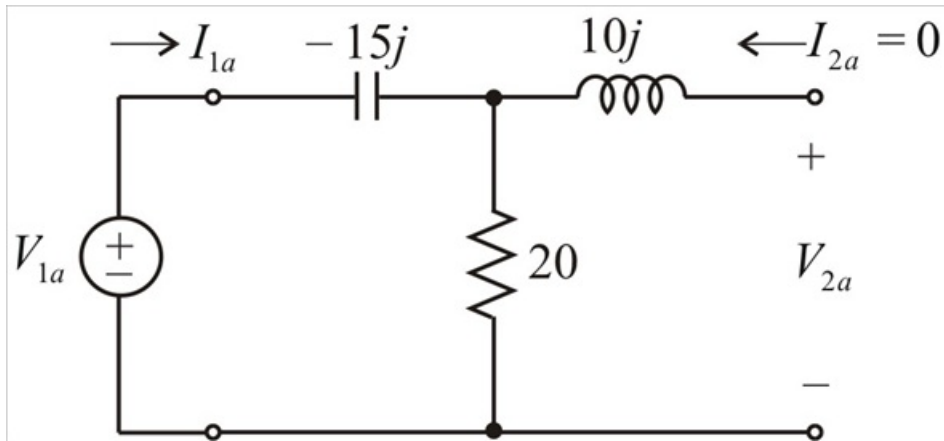


Step 2 of 15

To obtain Z_{11a} and Z_{21a}

Consider a voltage source V_{1a} to the input port and leave the output port open

Step 3 of 15



Step 4 of 15

Applying KVL, we get

$$V_{1a} = (-15j + 20)I_{1a}$$

$$\frac{V_{1a}}{I_{1a}} = (20 - 15j)\Omega$$

We know that

$$V_{1a} = Z_{11a}I_{1a} + Z_{12a}I_{2a}$$

$$V_{2a} = Z_{21a}I_{1a} + Z_{22a}I_{2a}$$

$$\text{Then } Z_{11a} = \left. \frac{V_{1a}}{I_{1a}} \right|_{I_{2a}=0}$$

$$\therefore Z_{11a} = (20 - 15j)\Omega$$

$$V_{2a} = 20I_{1a}$$

$$\frac{V_{2a}}{I_{1a}} = 20$$

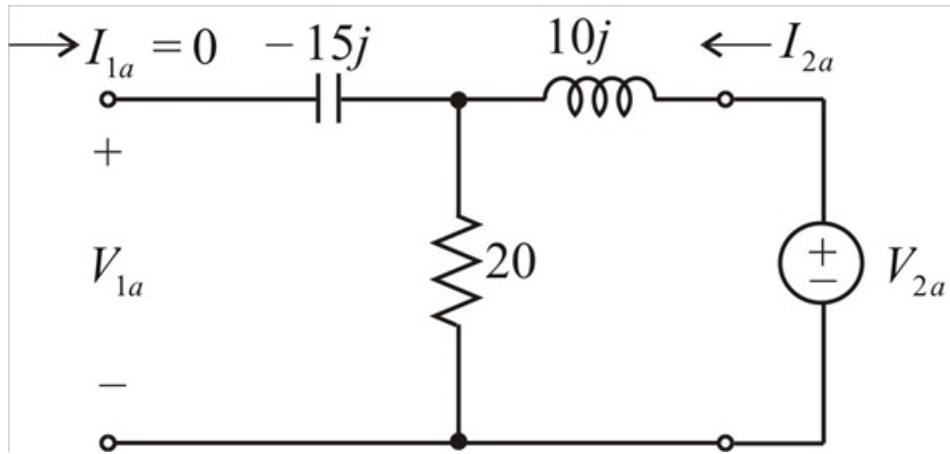
$$\text{When } Z_{21a} = \left. \frac{V_{2a}}{I_{1a}} \right|_{I_{2a}=0}$$

$$\therefore Z_{21a} = 20\Omega$$

Step 5 of 15

To obtain Z_{12a} and Z_{22a} consider a voltage source V_{2a} to the output port and leave the input port open

Step 6 of 15



Step 7 of 15

Using KVL, we get

$$V_{2a} = (20 + 10j)I_{2a}$$

$$\frac{V_{2a}}{I_{2a}} = 20 + 10j$$

When $Z_{22a} = \left. \frac{V_{2a}}{I_{2a}} \right|_{I_{1a}=0}$

$$\therefore Z_{22a} = (20 + 10j)\Omega$$

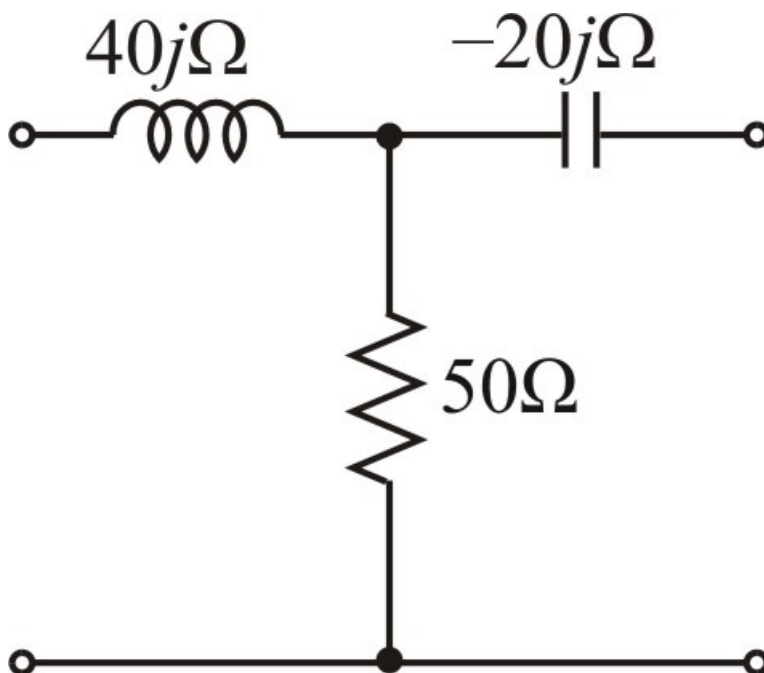
$$V_{1a} = 20I_{2a}$$

$$\frac{V_{1a}}{I_{2a}} = 20$$

When $Z_{12a} = \left. \frac{V_{1a}}{I_{2a}} \right|_{I_{1a}=0}$

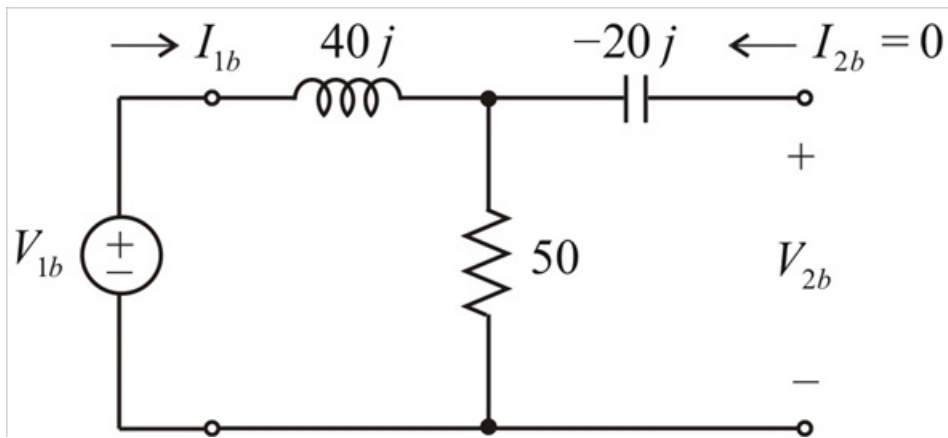
$$\therefore Z_{12a} = 20\Omega$$

For N_b :-



To obtain Z_{11b} and Z_{21b} consider a voltage source V_{1b} to the input port and leave the output port open.

Step 8 of 15



Step 9 of 15

$$V_{1b} = (50 + 40j)I_{1b}$$

$$\frac{V_{1b}}{I_{1b}} = (50 + 40j)\Omega$$

When $Z_{11b} = \left. \frac{V_{1b}}{I_{1b}} \right|_{I_{2b}=0}$

$$\therefore Z_{11b} = (50 + 40j)\Omega$$

$$V_{2b} = 50I_{1b}$$

$$\frac{V_{2b}}{I_{1b}} = 50$$

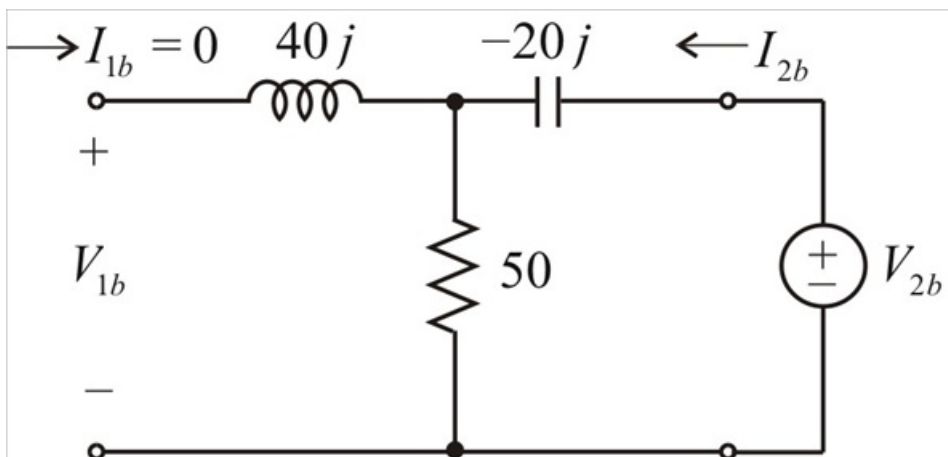
When $Z_{21b} = \frac{V_{2b}}{I_{1b}}$

$$\therefore Z_{21b} = 50\Omega$$

Step 10 of 15

To obtain Z_{12b} and Z_{22} consider a voltage source V_{2b} to the output port and leave the input port open

Step 11 of 15



Step 12 of 15

$$V_{2b} = (50 - 20j)I_{2b}$$

$$\frac{V_{2b}}{I_{2b}} = 50 - 20j$$

When

$$Z_{22b} = \left. \frac{V_{2b}}{I_{2b}} \right|_{I_{1b}=0}$$

$$\therefore Z_{22b} = (50 - 20j)\Omega$$

$$V_{1b} = 50I_{2b}$$

$$\frac{V_{1b}}{I_{2b}} = 50$$

When $Z_{12b} = \left. \frac{V_{1b}}{I_{2b}} \right|_{I_{1b}=0}$

$$\therefore Z_{12b} = 50\Omega$$

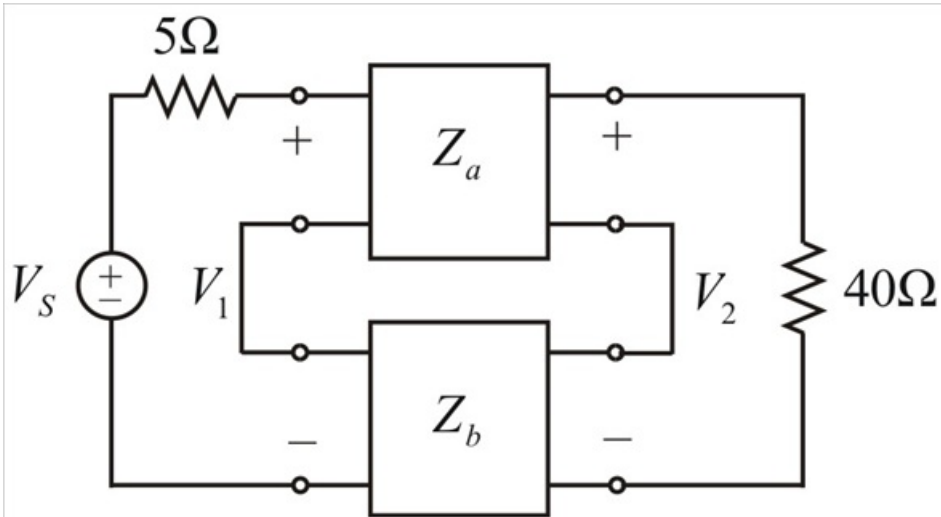
Step 13 of 15

Thus, $[Z] = [Z_a] + [Z_b]$

$$[Z] = \begin{bmatrix} 20 - 15j & 20 \\ 20 & 20 + 10j \end{bmatrix} + \begin{bmatrix} 50 + 40j & 50 \\ 50 & 50 - 20j \end{bmatrix}$$

$$\therefore [Z] = \begin{bmatrix} 70 + 25j & 70 \\ 70 & 70 - 10j \end{bmatrix}$$

Step 14 of 15



We know that

$$V_1 = Z_{11}I_1 + Z_{12}I_2$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2$$

$$V_1 = (70 + 25j)I_1 + 70I_2 \text{ ---- (1)}$$

$$V_2 = 70I_1 + (70 - 10j)I_2 \text{ ---- (2)}$$

At the input port,

$$V_1 = V_S - 5I_1 \text{ ---- (3)}$$

And at the output port

$$V_2 = -40I_2$$

$$I_2 = \frac{-V_2}{40} \text{ ---- (4)}$$

Substitute (4) in equation (2), we get

$$V_2 = 70I_1 + (70 - 10j)\left(\frac{-V_2}{40}\right)$$

$$V_2\left(\frac{11 - j}{4}\right) = 70I_1$$

$$I_1 = \left(\frac{11 - j}{280}\right)V_2 \text{ ---- (5)}$$

Substituting (3), (4) and (5) in equation (1), then

$$V_s - 5I_1 = (70 + 25j)I_1 - \frac{70}{40}V_2$$

$$V_s = (70 + 25j + 5)\left(\frac{11-j}{280}\right)V_2 - \frac{70}{40}V_2$$

$$V_s = \left[\frac{(75 + 25j)(11-j) - 70 \times 7}{280}\right]V_2$$

$$\frac{V_s}{V_2} = \frac{850 + 200j - 490}{280}$$

$$\frac{V_2}{V_s} = \frac{280}{360 + 200j}$$

$$\therefore \frac{V_2}{V_s} = 0.594 - 0.33j$$

$$\therefore \boxed{\frac{V_2}{V_s} = 0.6799 \angle -29.05^\circ}$$